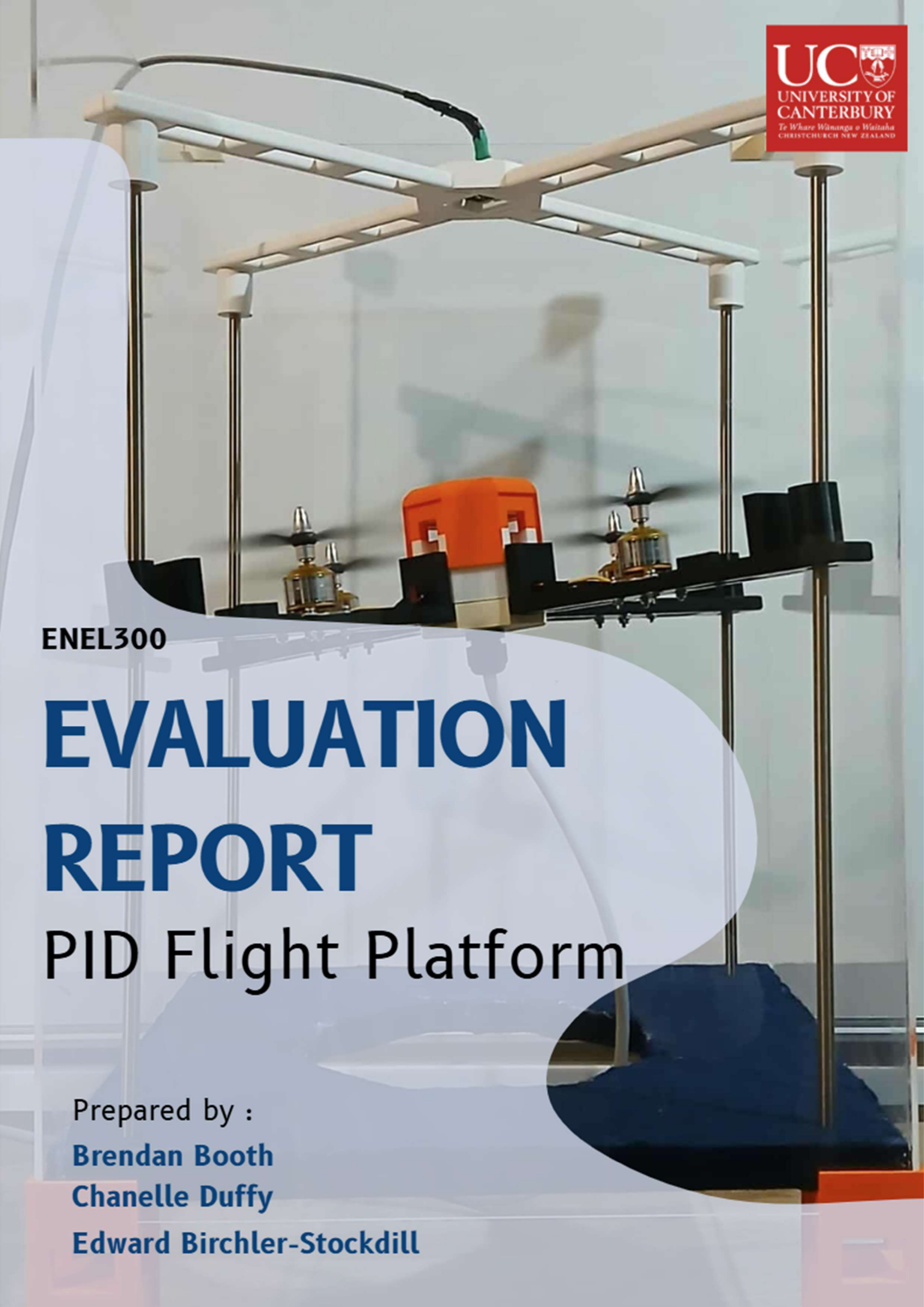

UC
UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

ENEL300

EVALUATION REPORT

PID Flight Platform

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1 SUMMARY

Seventeen requirements were identified for the project (R1 to R17). Of these, ten were categorised as essential to project success. Eight of these essential requirements were met by the prototype as demonstrated on 18/9/25. Of the two that remained unmet, R1 relating to project budget was superseded and no longer relevant to project success. The final unmet essential requirement, R6, related to controller behaviour identification and was assessed as partially met during demonstration. The capability of the prototype to fully meet R6 was clearly demonstrated, and the design is not limiting the fulfilment of this requirement in any way.

The remaining seven requirements were classified as desirable or optional and represent extension goals for the project, such as additional features or functionality, and were not deemed critical for project success. Two were not implemented for the project demonstration. Of these, R17 relating to in-flight changing of control constants was deemed unnecessary as it would have added very little to prototype functionality, and R9, relating to number of distinct control constant values is trivial to implement within the existing code base.

Thus, the project is deemed to be successful, and achieves the stated goal:

To design, develop and test a safe and portable device that physically demonstrates PID control system dynamics, using an engaging flight platform”

Prior to final prototype testing and demonstration, quantitative testing and assessment on subsystems was carried out on separate elements of the design to confirm function and performance. Battery runtime testing resulted in the use of a higher capacity battery to meet runtime requirements, sensor testing resulted in selection of a more accurate, faster ToF sensor, cost analysis resulting in securing further funding from primary stakeholder, and physical dimensions of prototype were modified to meet R3, R12, and R14.

Once a functional prototype was complete, performance testing was carried to tune the overall system to meet the overarching project goal of demonstrating PID control dynamics. This led to control system refinements and subtle changes to R9 and R17 as mentioned earlier, with associated justification and rationale.

A successful prototype demonstration was performed on 18/9/25. All changes to initial project requirements are justified and the team is satisfied with the delivery of the project. The design shows promise for extension work, and the primary stakeholder has expressed satisfaction with the outcome of the project.

2 RESULTS

This section provides high-level assessment of prototype against initial requirements.

2.1 EVALUATION AGAINST REQUIREMENTS SUMMARY

Table 1 lists all requirements identified at proposal stage and gives high-level requirement fulfilment status and brief justification. For requirements not explicitly met, more detail and explanation are given below. Subsequent sections in this report provide evidence of requirement fulfilment for more technical requirements.

Table 1 - Requirement fulfilment status

No.	Requirement	Fulfilment status	Justification	Priority
1	Total cost of build under $70	Not met	Requirement superseded as stakeholder funding secured (Section 2.2.5)	Desirable
2	Flight platform is constrained to the z-axis position.	Met	The flight platform is constrained by four poles – one for each arm.	Essential
3	Flight platform range of motion greater than 300mm	Met	The flight platform is constrained vertically to a 490mm range, the motion spans this whole range. (Section 2.3.1)	Essential
4	A PID controller is controlling the movement of the flying platform.	Met	The motors are controlled via a software implemented PID controller. (Section 2.3.1)	Essential
5	The control constants each have three preset values including zero	Met	Low, Medium, High values for each control constant implemented in controller. (Section 2.3.1)	Essential
6	Variation in control dynamics between different controller combinations	Partially Met	During demonstration, stakeholder identified three distinct responses. Presets to be improved to meet this fully. (Section 2.3.1)	Essential
7	Control constants are changed via a dedicated user interface.	Met	Control constants are changed using three potentiometers	Desirable
8	Control constants adjustable between flights within 1 minute	Met	Control constants can be changed easily between flights by adjusting the potentiometers	Essential

9	The control constants each have six preset values including zero	Not met	Further tuning of presets will enable prototype to meet this requirement	Optional
10	Flight platform sustains flight for at least 10 seconds	Met	Successful tests of duration >1m completed. Calculation suggests sustained flight times >1 hour	Essential
11	The flight platform is enclosed to prevent contact with any moving or electrically active components.	Met	All electrical connections in junction boxes and insulated with heat shrink. Platform enclosed with transparent acrylic. (Section 2.2.3)	Essential
12	Conforms to all University safety regulations.	Met	Lecturer deems prototype suitable for lecture demonstration	Essential
13	Flight platform enclosure does not hinder visual observation.	Met	The platform is enclosed in transparent acrylic. Very little visual distortion introduced. (Section 2.2.3)	Essential
14	Transportable to and from lecture theatre	Met	It can be carried or transported on a trolley by one person. (Section 2.2.4)	Desirable
15	Flight platform responds to external disturbance applied by user.	Met	The flight platform can be disturbed through the power cable. The system responds to this disturbance based on the controller implemented. (Section 2.3.3)	Optional
16	Ground effect, turbulence and frame should not cause the flight platform's position to deviate by more than ±100 mm from the settled point.	Met	Once settled the drone will not deviate from its position. See section 2.2.7 for evidence of satisfaction of this requirement	Desirable
17	Control constants can be changed while the platform is in flight.	Not met	If the need arises this can be implemented. However, for the current implementations it was deemed unnecessary.	Optional

2.1.1 Outstanding requirements

Requirement 1 - Total cost of build under $70

After submission of proposal document in which this requirement was given, verbal assent to exceed the base project budget was given, based on the promise demonstrated by the project. Full cost analysis is provided in section 2.2.5. Primary stakeholder has agreed to provide additional funding

Requirement 6 - Variation in control dynamics between different controller combinations

The success criterion for this requirement was defined as: In a randomised demo of 5 presets, the primary stakeholder correctly identifies relative response differences in ≥ 4/5 cases.

At the project demonstration, the primary stakeholder recognised the three presets which had been tuned and prepared for the demonstration. Time constraints precluded the addition of the two presets required to meet this requirement, but subsequent development has resulted in the fulfilment of this requirement.

Requirement 9 - The control constants each have six preset values including zero

Preset values of control constants are used to very controller behaviour for the demonstration. Due to time constraints, an effective, but compact, selection of three different values for each control constant was used. This met *essential* requirement 5 (The control constants each have three preset values including zero, but precluded satisfaction of this *optional* requirement, which was conceived as an extension of requirement 5 anyway. To meet requirement 9, only a simple change in the code base is required.

Requirement 17 - Control constants can be changed while the platform is in flight

This requirement was deemed unnecessary for project demonstration. The rationale behind the requirement originally was that the demonstration efficiency would be improved by limiting the length of time the flight platform was stationary for. This turned out to be inconsequential as the flight platform can be stopped, the control constants adjusted and then started again in a matter of seconds if required. Furthermore, the prototype is quite noisy during operation, and periods of relative silence are quite useful to explain what is being demonstrated. However, as with requirement 9, only a simple change in software is required to implement this if deemed necessary later.

2.2 TESTING AND FUNCTIONALITY OF SUBSYSTEMS

2.2.1 Sensor Testing

As explained in the design report a ToF sensor was decided on over an IR or ultra-sonic due to its higher sampling frequency and range of accuracy. From the sensors available, the VL53L1X ToF sensor was chosen, this sensor was capable of a maximum sampling rate of 50Hz, at this sampling rate the range of the sensor is reportedly 4-135 cm under strong ambient light [1]. To ensure the sensor would work for our system, it was tested under the following conditions:

- In bright daylight.
- At two different sampling frequencies; maximum (50Hz) and 40Hz.
- Sensor readings were filtered (Median of 3, EMA).

- All measurements listed below are the median of ~30 samples due to the +/- 5 mm variation post filtering.

Table 2: Results of ToF sensor at 40Hz and 50 Hz sampling frequencies.

Ruler (mm)	Sensor at 40Hz (mm)	Sensor at 50Hz (mm)
410	488	Out of Range
350	348	Out of Range
300	301	Out of Range
250	250	255
200	201	200
150	149	147
100	100	98
50	53	50
20	19	20

From this testing, it was determined that when running at its maximum sampling frequency, the ToF sensor did not reliably measure the range required. The sensor results when running at 40Hz were far more accurate.

It should be noted that the first measurement at 40 Hz is 488mm when the flying platform is sitting at 410mm. This anomaly is due to the sensor reading the base height (489mm) instead of the platform's height. In future it would be prudent to use a sensor that has a narrower cone of vision to prevent this issue.

Overall, this sensor running at 40 Hz was determined to have a reasonable level of accuracy to demonstrate the prototype.

2.2.2 Battery runtime testing

The battery runtime is critical to requirement 10, and other aspects of the selected battery have direct relevance to requirements 12, and 14. As the prototype's application space and use cases were further explored, an additional requirement was identified: the battery must support repeated flights within a single demonstration period.

Testing with the motors outside of the flight platform using a 10A DC laboratory supply was carried out to establish baseline design specifications and confirm datasheet values. These are given in Table 3.

Table 3 - Baseline current draw

	No-load Single motor	No-load Two motors
Current at steady state	0.58A	1.15A
Peak observed current	4.7A	>10A

During early project development, three 1.2Ah SLA batteries in parallel where used. However, these batteries depleted rapidly during testing, and it was noticed that as the state of charge decreased, motor dynamics become more subdued due to the lower voltage and current supply capacity.

Additional testing and calculations with a larger battery were carried out to determine the load profile of the motors and hover platform. At this point, the motor and platform were finalised, and this testing would provide values to match expected real-world operation. A 12V automotive starting 650CCA, 65Ah,110RC battery was used and was carried forward into the final prototype. Results are given in Table 4.

Table 4 - Total prototype operating current draw

	Current
Current at steady state hover	14.8A
Peak observed current during max power	47A
Transient inrush current peak (Using oscilloscope)	~75A

With these results and battery specifications, estimations for total prototype runtime can be derived. Battery specifications and effect on prototype operation are given in Table 5

Table 5 - Battery specifications and effects

Specification	**Description**	**Effect**
CCA: 650 (Cold-cranking amps)	Rated to deliver 650A for 30 seconds at -18°C while maintaining voltage above 7.4V.	Battery easily able to meet peak current requirements of prototype
Ah: 65 (Amp-hours)	Energy storage capacity. Theoretically able to deliver 65A continuously for one hour (real-world limitations ignored)	Theoretically - assuming hover and complete discharge - prototype can continuously operate for 4.4hours.
RC: 110 (Reserve capacity)	Rated to deliver 25A at 27°C for 110 minutes before voltage drops below 10.5V. more relevant to real-world applications	Since hover current is less than 25A, continuous operation of >110 minutes expected.

These results indicate that continuous operation for an open day, faculty demonstration, or lecture of at least 1 hour is achievable with considerable safety factor. Limitations with these estimations include:

- Prototype unlikely to be operated continuously.
- Assumption that hover current is average current.
- Assumption that effect of higher short-duration currents will not have significant effect on continuous ratings

2.2.3 Acrylic enclosure

R11 and R12, which specified the need for an enclosure to ensure safety, were met using 3 mm clear acrylic. A laser cutter was used to produce clean cuts, resulting in a precise and professional finish. R13, that the enclosure must not hinder visual observation, was also satisfied by ensuring clear visibility of the flight platform from all angles.

The flight platform is fully enclosed to prevent contact with moving or electrically active components and complies with all University safety regulations.

2.2.4 Sizing

Requirement 14 defined that the design should be transportable to and from lecture theatres. The prototype has the following sizing specifications.

- 410x410x700mm (Width/length/height)
- 7.7 Kg (Battery excluded)

Based on these measurements it was determined that the design achieved R14 as it can be carried by one person (excluding the battery).

2.2.5 Cost Analysis

One of the requirements that was not met was R1, however the primary stakeholder has agreed to provide additional funding. Table 6 below outlines the total cost of the project. For the prototype a car battery was used (see section 2.2.2), this battery was excessive for what was required. For this reason, the battery has been left off the cost analysis, further research is required to find a suitable battery for this project.

Table 6: Costs breakdown.

Parts	Cost (per unit)	Supplied from	Number of units
Motors + ESC's + Props	$90.00	Dick Smith	1
ToF Sensor	$10.00	UC	1
Arduino Shield	$18.90	Jaycar	1
Logic level converter	$2.50	UC	1
Arduino Uno R3	$50.00	Digikey	1
Cables + Glands + Junctions	$20.00	Jaycar	1
Potentiometers	$1.20	UC	3
Button	$2.50	UC	1
Acrylic casing	$37.00	Mitre10	2
3D printed parts	$0.00	UC	-
Total	**$271.50**		

2.2.6 Hardware

The initial weight of the platform was 671.1 g, as it was built for strength. The design also included prop guards, which were later deemed unnecessary due to the protective enclosure and subsequently removed. The system's weight was further reduced to 421 g (excluding motors), making it more responsive. These adjustments helped meet requirements R3 and R10.

The platform originally used linear bearings to move along the guide rods. These bearings caused the system to pinch on the rods, so a new solution was required. The revised design incorporated wider guide rod sleeves, which eliminated pinching while allowing for a small amount of pitch and roll.

Initially, the base sat directly on the ground, with the cable running across the top. Because the cable was stiff, it sometimes tilted the drone, leading to unreliable dynamics. To address this, and to better satisfy requirements R15 and R16, a hole was cut in the centre of the base, which was then raised 100 mm on legs. This allowed the cable to hang freely without interference.

2.2.7 PID Response

To confirm that the control system was giving the desired response, the sensor height measurements were plotted against time for the core controller types. In this project, the goal is to have easily recognisable control dynamics. The Figures below demonstrate the extremes of the different controllers.

The ensuing sections show measurable proof of meeting the following requirements:

- **R3** – As demonstrated by the figures below, the range of movement of the flying platform is over 300mm. In the case of the integral controllers, the whole range is utilised to show control dynamics.
- **R4** – The responses plotted below clearly demonstrate the use of a software PID controller. The plots are what is to be expected for the different combinations of P, I and D, given the system and gain values.
- **R5** – The figures show that there are three preset values including zero. This allows for a variety of control responses ranging from pure proportional control to combinations of controllers such as PI and PD.
- **R6** – Based on our acceptance condition, this requirement was not met. Due to time constraints only four controllers were tuned to the point where their response was recognisable. However, of the controllers listed below, three out of four were able to be recognised by our stakeholder Chris Hann.

Proportional

The proportional controller was designed to demonstrate the large overshoot and oscillations that occur with P control. The response of the flight platform when set with a gain value of $K_p = 0.18$ is demonstrated in Figure 1 below. The initial overshoot is clearly visible, as are the following oscillations that gradually reduce. At steady state the proportional controller will still have small oscillations as it can only reduce steady state error, it cannot eliminate it.

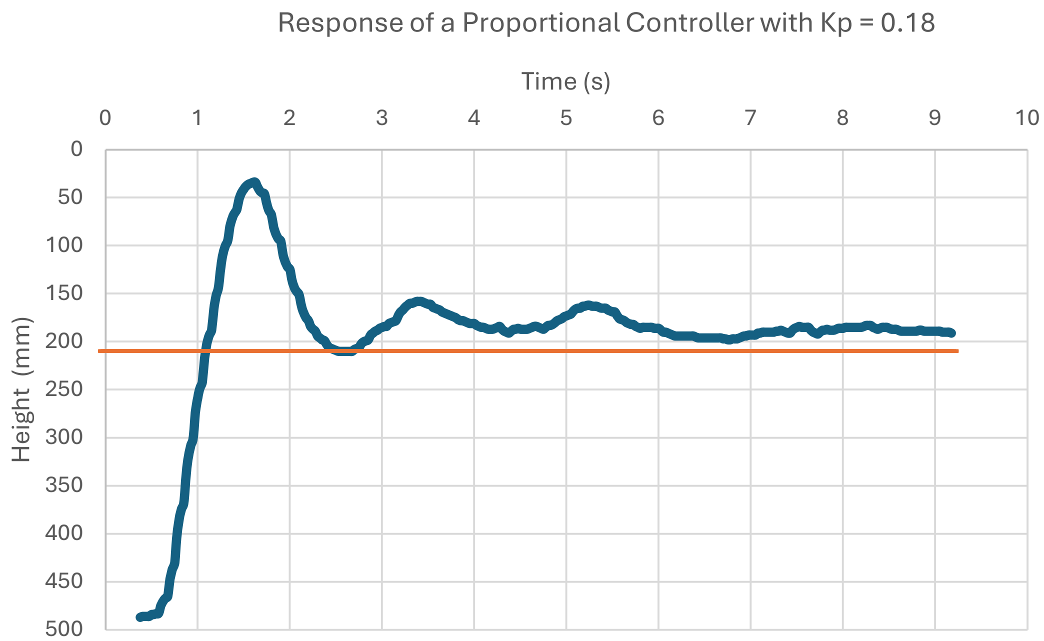


Figure 1: Height vs time for a proportional controller.

Integral

The integral controller was tuned to demonstrate the slow accumulation of error at the start, followed by the overshoot caused by the integral windup and the subsequent overcorrection before settling. In theory, an integral controller can remove steady-state error over time. However, in practice, real-world disturbances make this unlikely. These recognisable responses can be seen in Figure 2.

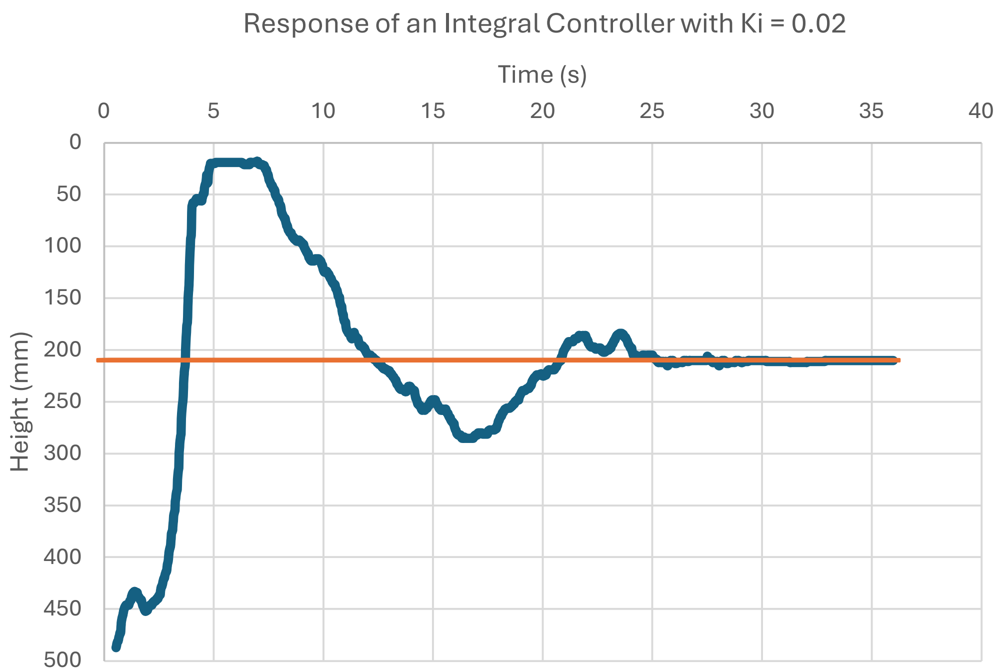


Figure 2: Height vs Time for an Integral controller.

Proportional-Derivative

The PD controller was designed to show the damping effect that the derivative component has on proportional control. To show the extreme effects of the derivative component it was tuned to demonstrate an over-damped system. The flight platforms response to a $K_p = 0.15$ and $K_d = 0.05$ can be seen in Figure 3 below.

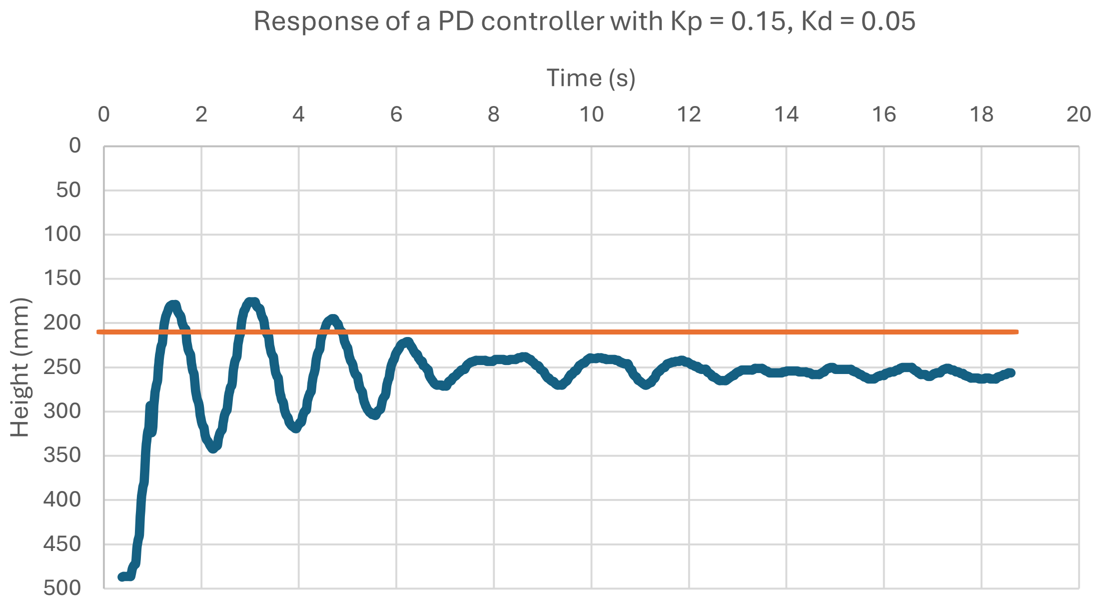


Figure 3: Height vs Time for a Proportional-Derivative controller.

This figure shows a heavily damped response, there is very little initial overshoot, however the oscillations are not as damped as they should be. This is likely due to a mixture of noise- which the derivative component amplifies- and poor tuning. Given more time, further tuning of this PD controller would be prudent.

This controller also causes the flying platform to settle slightly off of the setpoint, this is the expected behaviour for an overdamped PD controller as there is no integral component to correct the steady-state response.

Proportional-Integral

The sought after PI response was to demonstrate the quick rise time of proportional control and the ensuing oscillations before settling on the setpoint. Figure 4 below shows the response of a PI controller with a $K_p = 0.15$ and $K_i = 0.01$. This controller demonstrates the extreme overshoot and oscillations from the proportional component before slowly converging to the setpoint due to the integral component.

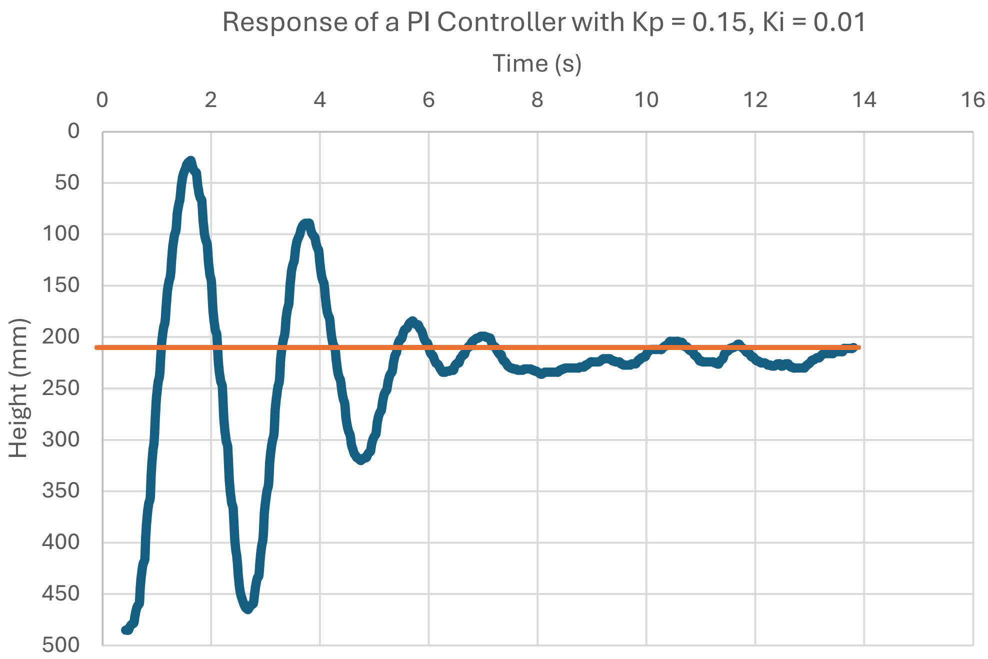


Figure 4: Height vs Time for a Proportional-Integral controller.

2.2.8 Steady State Testing

To ensure consistent long-term operation of the flight platform, three trials were conducted for the four core controller types. These trials consisted of 100 samples per controller, taken after the flight platform had been airborne for over 30 seconds. Table 7Table 1 below contains the mean error, this is calculated to show the average difference between the measurements and the setpoint (210 mm). The standard deviation demonstrates the consistency between trial runs. Finally, the max error shows the maximum amount the flying platform moved from the setpoint.

Table 7: Results of three trials with 100 samples each.

Controller	Trial Num	Mean error (mm)	Std Deviation (mm)	Max Deviation (mm)
	1	41.23	3.30	+48
P	2	32.75	4.74	+41
	3	-15.31	13.14	-35
	1	3.78	3.61	+13
I	2	-4.81	12.62	+23
	3	5.25	5.51	+15
	1	30.17	7.50	+40
PD	2	-11.52	7.97	-26
	3	7.69	8.50	+21
	1	32.44	3.60	+38
PI	2	2.44	10.47	+23
	3	6.48	4.08	+12

These results prove that the prototype has met requirement 16, none of the controllers at steady state vary in height more than +/- 100mm across all three trials. This data also demonstrates the effect the different types of controllers have on steady state operation. As expected, the controllers with a proportional component but no integral component never fully settle, this is evident by the larger mean error values. Controllers containing integral components have a much lower mean error.

A summary table can be found below (Table 8) showing the consolidated results for each controller. This data shows the consistency between trials, with a maximum standard deviation of 5.21 mm.

Table 8: Consolidated trial data.

Controller	Mean error (mm)	Std Deviation (mm)	Max Deviation (mm)
P	19.56	5.31	+48
I	1.41	4.75	+23
PD	8.78	0.50	+40
PI	13.79	3.83	+38

2.2.9 Disturbance

Requirement 15 outlined the optional extra of having the flying platform respond to external disturbances applied by the user. The design met this requirement through the high integral controller setting with $K_i = 0.02$.

By holding the flying platform down at the base via the power cable, the effects of integral wind-up can be demonstrated. The control output for this experiment is plotted against time in Figure 5, the measurements were taken from the point when the flying platform touched the base.

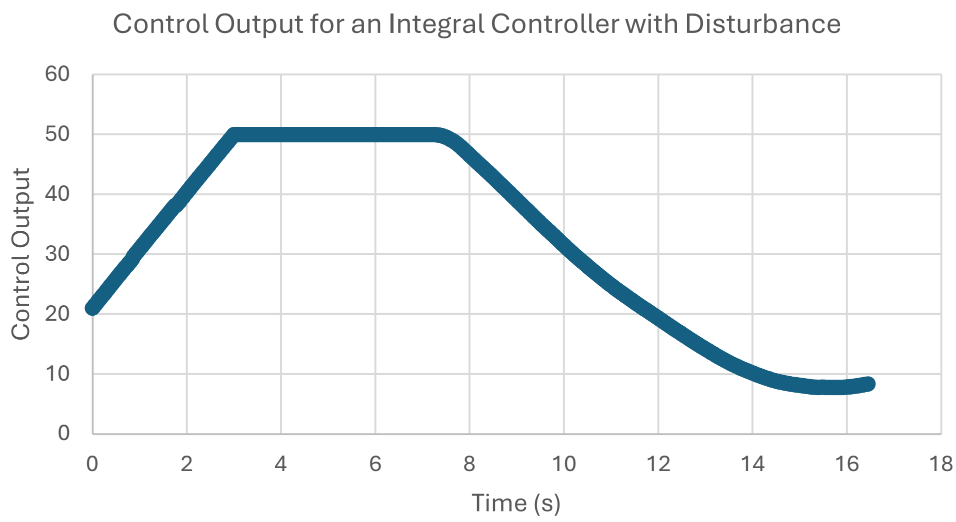


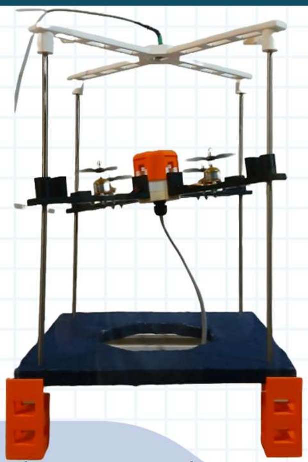
Figure 5: Control output when an external disturbance is applied on an Integral controller.

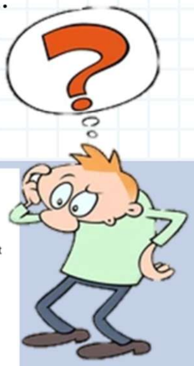
As demonstrated in the figure, the controller output slowly ramps up as it accumulates error to try and reach the setpoint. The output plateaus at the maximum integrator limit (50), at this point the flight platform is still being restrained by the user. The cord is then released, resulting in overshoot due to the large, accumulated error. This is shown on the graph as a gradual decrease in the control output back to zero.

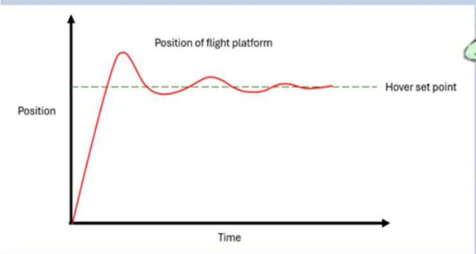
3 VISUALIZE YOUR PROJECT

CONTROL SYSTEMS
PID FLIGHT PLATFORM

What is a control system?
A control system is a setup that automatically keeps something working the way we want it to, by constantly checking what's happening and making small adjustments when needed- such as cruise control.







Problem: Students struggle wih linking conceptual theory of control systems with real world system dynamics

THE SOLUTION...

Build a hover platform that clearly demonstrates how the most common controller (Proportional, Integral, Derivative) automatically fine-tunes motion to affect stability, balance, and responsiveness in real-time.

FEATURES

- Test 9 control constant combinations to observe interactions.
- Introduce disturbances (e.g., pulling cable) to demonstrate integral windup.
- Up to an hour runtime
- Beats following a line on a graph!

4 FUTURE WORK AND CONCLUSION

4.1 TESTING OUTCOMES

The main results from subsystem testing and overall prototype performance testing are:

- The current battery is expected to be capable of powering the prototype for over an hour.
- The current ToF sensor required sample speed to be reduced to maintain measurement accuracy.
- The complete prototype successfully exhibits desired controller behaviour, but is limited by sensor sampling speed.
- The design is capable of meeting all extension requirements unmet as of project demonstration with trivial modification to code base.

4.2 FUTURE WORK

The design is already at a workable stage, as demonstrated by the sections above. However there are a number of future improvements that could be implemented to build upon and improve the current design.

- A custom-made control panel in place of the Sistema container currently serving as the control panel. 3D printing or laser cutting an enclosure from opaque material, labelling all user inputs and controls, and improving the mounting of the hardware within would not require much further work or expense for an impactful effect of prototype appearance.
- Sistema container underneath base replaced with dedicated electrical junction box, or alternatively, a male-female plug and socket arrangement.
- Substitute existing PVC sheathed 6 core control cable to flight platform with similar silicon sheathed cable. This type of cable is far more flexible and will have less impact on system dynamics [2].
- Finding a more suitable sensor would greatly improve the control response of the system. The experiments above have proven that a sensor with a greater maximum sampling rate, a narrower cone of vision, and less noise would go a long way in improving the control response.
- A downsized battery is recommended—the present unit far surpasses the operational load profile.
- Several of the control responses require further tuning, as seen in section 2.2.7. Given more time, it would be helpful to have a well-tuned PID option along with the other controller combinations.
- The addition of a live graphical control feed would further enhance the project's value as a teaching aid.

4.3 POTENTIAL HAZARDS

Safety to the standard required for lecture demonstration was identified as a key design requirement and the design reflects this, with the inclusion of acrylic shield to prevent accidental contact by the user or observers with the flight platform.

The application space for the design assumes a competent user demonstrating to observers some distance away. The design is not entirely enclosed, with openings top and bottom to allow for air movement and tether cable and sensor. This means intentional contact with the flight platform is possible. Therefore, the user should supervise and take due care to inform unfamiliar observers of the possibility of harm should they try to reach into the top or bottom and touch the flight platform. This is not likely, but still possible.

Standard procedures for safe handling and use of SLA batteries must be followed [1]. The terminals of the battery are exposed during demonstration set up and dismantling, and this presents the risk of inadvertent short circuit. The user should ensure that anyone assisting is aware of this.

5 REFERENCES

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